Hallmarking

Hallmarking Module - Functional Specification

1. What this system does

This system manages the **complete hallmarking process** — from item pickup to final return.

Each order moves step-by-step through different desks.
You only work on **your assigned desk**, complete your part, and move it forward.

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2. Basic Flow

1. Items are picked up from jeweller
2. Hallmarking request is created
3. Items are tested (XRF → Sampling → Fire Assay / Silver Titration)
4. Passed items are marked (HUID)
5. Final weight is recorded
6. Items are returned and billed

SNO	BLOCK	FUNCTION
1	Delivery & Pickup Desk	deals with pickup of incoming items for the hallmarking process and delivery of items whose hallmarking has been completed.
2	Hallmarking Requests Desk	A Hallmarking request is created here by entering all item details and then passed to the quality manager desk.
3	Quality Manager Desk	The items received from the hallmarking request are sent to XRF Testing from here.
4	XRF Testing Desk	The XRF testing results are entered for each piece in items here. Each item is given a pass or fail verdict in this state based on claimed purity.
5	Sampling Desk	The items passing the XRF Test are then sampled and sampling details are entered here. Only XRF passed items can be sent to fireassay here.
6	Fireassay Desk	Fire-assay test results are entered for each item and a pass or fail verdict is given.
7	Silver Titration Desk	The person at the Silver Titration desk will see only the orders in Silver Titration state. Silver Titration test results are entered for each item and a pass or fail

		verdict is given.
8	HUID Printing	7.1) HUID marking : marking is done for each piece that has passed the Fire-assay test. 7.2) At the end of each day a count of total markings done for that day needs to be sent to the laser marking module for validation. This is done in 2 ways: 1. Periodic : at 10 PM each day the count is sent automatically. 2. Manual : the user closing a day's markings can send the count manually to the laser module using the Send HUID count menu under the HUID Printing desk.
9	Final Weighting	All the items along with samples in the HM request are weighted final time before returning to the customer to analyse weight reductions due to testing. Here all items should appear even though some items can be rejected for intermediate processes in the HM process.
10	Billing	All the items are considered irrespective of rejections for some tests for bill The charge is an overall amount based on the number of items in one HM request.

3. Functionality

1. **Delivery Desk**
   A. Orders
      1. All orders
         - Upon opening this section, the user can view all orders- pickup and return.
         - User can create a new order by giving customer's name and further details about them and their items and choose a delivery type(delivery pickup or delivery return)
         - After entering some required details, user can click on 'Send for Pickup' and the order moves to "awaiting pickup' stage where user has to enter Customer gross and Customer net weights in grams, as well as upload images of -Items on Weighing Machine and Jeweller Receipt and then clicks on 'Mark as picked up'
         - Then the user has to enter PHC quantity, PHC Gross Weight and Declared Purity and clicks on 'Mark as Received'.

      2. Incoming items

- Here, the user can view all Incoming items(items in transit) and mark them as received, after he enters PHC quantity, PHC Gross Weight and Declared Purity for the items in the order.
- Users can also create new orders.

3. Returns
    - Here, there are orders that the customer wants delivered back to his place after being tested and 'Mark as Delivery' after entering necessary details.
    - Users can also create new orders.

B. My Tasks
1. My Pickups
    - Here, orders which are supposed to be picked up are displayed and can be viewed by the user and delivery person.
    - Once orders move to the 'Received' stage, they are sent to the hallmarking desk and can be viewed there.
    - Users can also create new orders.

2. Hallmarking Desk

A. Hallmarking Requests
1. Here, there are orders which come from the pickup desk as well as on the spot orders which can be created by users.
2. In a hallmarking request order, the user has to enter BIS number, among other details and can send it to the next stage, Quality Manager.

B. Quality Manager Desk
1. Here, users can generate Tag IDs for the items and then send them to the next stage, XRF Testing.

C. XRF Testing
1. Here, the user has to enter the XRF test result for each item, and results are instantly calculated showing whether the item passed or failed the test.
2. After performing the test, the user has the option to abort the process and skip to the final step in hallmarking, which is final weighting. There is a button on top for the same, next to the button which moves the order to the next stage, Sampling Desk.

D. Sampling Desk
1. Here, items which passed the XRF test can be selected for sampling for Fire assay, and records are sent accordingly.

E. Fire Assay

1. Here for items that have passed XRF, you can enter Fire Assay results, and the result is calculated, showing if you have passed or failed the test.
2. After performing the test, similar to XRF testing, the user has the option to abort the process and skip to the final step in hallmarking, which is final weighting. There is a button on top for the same, next to the button which moves the order to the next stage, HUID printing.

3. **HUID Printing Desk**

A. HUID Marking
   1. Orders come here after the Fire Assay test and users can click on Start marking.
   2. User can also click on the checkbox 'Remarking' if he wants to do the same.
   3. After Start marking, you can move to Final Weighting.

B. Final Weighting
   1. Here the user has to enter the final weight and for all the items after performing the tests.
   2. After that, the user can move to the completed stage.

C. Send Daily HUID Count
   1. Here, users can see the total number of markings done per day.
   2. There is a scheduled process, which is triggered daily at 10pm, to send this count to the Laser Marking module. Users can do this manually as well by clicking on the button.

4. FORMS

1. Delivery Order Form

Mandatory:

- Customer Name
- Delivery Type (Pickup / Return)
- Date

Key fields:

- Quantity
- Customer Weight
- PHC Weight
- Declared Purity

2. Hallmarking Request Form

Mandatory:

- BIS Request Number
- Customer Name
- Material (Gold/Silver)
- AHC Center

Item Line Mandatory:

- Item Category
- Declared Quantity
- Declared Weight

3. Quality Manager Form

Mandatory:

- Job Number
- Tag ID

4. XRF Testing Form

Mandatory:

- Upload .exp file
- Tested Karat

System auto:

- Purity %
- Pass / Fail

5. Sampling Form

Mandatory:

- Select items for Fire Assay (only passed ones)

6. Fire Assay / Silver Titration

Mandatory (basic):

- Sample details
- Measurement values

System auto:

- Purity
- Pass / Fail

7. HUID Marking Form

For passed items only:

- Marking Date
- Total Markings

8. Final Weighting Form

Mandatory:

- Final Weight (for ALL items)

5. Status Reference

Each record in the system has a Status field shown in the status bar at the top of the form. The table below explains what each status means for the HM Request (the central record).

Status	**What it means**
Draft	The HM Request has been created. Item details and weights are being entered. Still editable.
At Quality Manager	Moved to the Quality Manager Desk for job number and tag ID assignment.
XRF Testing	XRF Testing record created; waiting for scan results to be entered.
XRF Completed	XRF results entered and confirmed; waiting to move to Sampling.
At Sampling Desk	Sampling record created; operator is selecting items for fire assay.
Fire Assay	Fire Assay record created; fire assay measurements being entered.

HUID Printing	Fire assay completed; items are being marked with HUID.
Completed	Final weighting done; the order is fully closed.
Cancelled	The order has been cancelled. No further action is possible.

Delivery Order Statuses

Status	**What it means**
Draft	Order created; still being edited by HC/Admin staff.
Awaiting Pickup	Sent for pickup; delivery person can see this in My Pickups.
In Transit	Delivery person has collected the items; AHC is waiting to receive them.
Received	Items received at AHC; PHC weights and purity entered. Order complete.
In Transit (return)	Items handed to delivery person for return to jeweller.
Delivered	Items delivered back to the jeweller. Return order complete.

6. Common Questions

Can I create a Quality Manager or XRF record manually?

No. Records at every stage from Quality Manager onward are created automatically by the system when you move an order forward. You only need to open the record and fill in the fields for your stage.

What happens when an item fails XRF testing?

The item is automatically recorded in the Rejected Items log on the HM Request. At the Sampling Desk, the failed item appears in the list but is greyed out and cannot be selected for fire assay. The item still appears at Final Weighting so its final weight can be recorded.

What does Abort Process do?

At the XRF Testing and Fire Assay stages, if one or more items have failed, the Abort Process button appears. Clicking it skips all remaining stages (Sampling, Fire Assay, Marking as applicable) and goes directly to Final Weighting. This is used when the jeweller does not want to continue with the remaining items.

What is the difference between Declared Weight and Observed Weight on the HM Request?

Declared Weight is what the jeweller claims when they hand over the items. Observed Weight is what the AHC physically measures when the items arrive. Ideally these should match within 0.5g. If they differ significantly, the Weight Status shows a Warning and the discrepancy should be investigated before proceeding.

How is the daily HUID count sent?

Every day at 10:00 PM, the system automatically calculates the total markings done that day and sends the count to the Laser Marking module. If you need to send it earlier (for example when closing for the day), go to HUID Printing > Send Daily HUID Count, verify the count, and click Send to Laser Marking.

Can I edit a record after it has moved to the next stage?

No. Once a record is moved to the next stage, the fields of the current record become read-only. You can always view historical data, but changes must be handled through appropriate corrections or cancellations.

Testing

Testing Module — Functional Specification

1. What This Module Does

- The Testing Module handles end-to-end processing of customer orders for metal purity testing.
- A customer walks in with gold, silver, or platinum items, and the module tracks those items through one or more lab tests — XRF, Fire Assay, and Silver Titration — until results are finalised and the order is ready for billing.
- Each order is called a **Testing Job**. A job can contain multiple items, and each item is assigned to a specific test based on the metal type.
- The job moves through test stages in sequence, and at each stage, the relevant lab operator enters results. Once all tests are complete, the job moves to billing and then to done.

Core Motto of the Module:

- This module is to enable the processing of orders of customers who need testing as a service only.
- The customers may come for XRF, Fireassay and Silver Titration.
- Each customer's testing job can include multiple tests- so the job/order passes through all required tests.
- The billing is done at the end

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2. Desks — Who Uses What

The module is organised into **desks**, each designed for a specific role. When a user opens their desk, they see only the jobs that are currently at their stage.

Desk	Who Uses It	What They See
Reception Desk	Receptionist	All jobs in Draft state — new orders waiting to be created or confirmed
XRF Desk	XRF Technician	Only jobs currently in the XRF testing stage
Fire Assay Desk	Fire Assay Technician	Only jobs currently in the Fire Assay testing stage

Silver Titration Desk	Titration Technician	Only jobs currently in the Silver Titration testing stage
All Jobs	Manager / Admin	Every job regardless of state — used for oversight and reporting

Each desk opens with a **Kanban view** (card-based), and can also be switched to List or Form view.

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3. Creating a New Testing Job

3.1 Starting a Job

The receptionist opens the **Reception Desk** and clicks **New**. A blank job form opens.

3.2 Filling in Job Details

The form has two main sections side by side:

Left — Service Details:

Field	What to Enter
Service	Pre-filled as "Testing Service"
Date	Auto-filled with current date and time
Branch	Select the branch where testing is happening
Book No	Physical book reference number (optional)
Serial Number	Serial reference (optional)

Right — Customer Information:

Field	What to Enter
Customer Name	Select from customer list (required). All other customer fields auto-fill
Customer Number	Auto-filled from customer record

Customer Company	Auto-filled
Customer GSTIN	Auto-filled
Customer BIS Number	Auto-filled
Customer Address	Auto-filled

Order ID appears in the sticky header bar at the top of the form. It is auto-generated once Customer and Branch are selected.

The format is: `BranchCode-TS-CustomerNumber-SequenceNumber` (e.g., `BLR1-TS-001-041`). This ID stays visible at all times as the user scrolls through the form.

3.3 Adding Items

Below the header sections, the **Service Items** tab contains the items table. The receptionist adds one row per item the customer has brought for testing.

Column	How It Works
S.No	Auto-numbered (1, 2, 3...). Updates automatically when rows are added or removed
Item Name	Select from the product list (e.g., "Ring", "Chain", "Bangle")
Type	Defaults to "Type 1". Can be changed to Type 2, 3, 4 etc. New types can be added
Metal	Select Gold, Silver, or Platinum. Defaults to Gold
Qty	Number of physical pieces for this item. Defaults to 1. For XRF testing, each piece gets its own reading
Weight Claimed	Weight the customer declares (in grams)
Weight Recorded	Weight measured in the lab (in grams)
Weight Check	Automatically shows a green "Match" badge if both weights are equal, or red "Mismatch" if different

Purity Claimed	Select from the purity list (e.g., "995", "999", "K22"). New values can be added by users
Test	Select which test this item needs: XRF, Fire Assay, or Silver Titration

Important rules about Metal and Test combinations:

- Gold items can only have XRF or Fire Assay
- Silver items can only have XRF or Silver Titration
- Platinum items can only have XRF
- If the user selects "Silver Titration" as the test, the Metal auto-changes to Silver
- If the user selects "Fire Assay" as the test, the Metal auto-changes to Gold
- Selecting an invalid combination shows a warning and clears the test selection

3.4 Confirming the Job

- Once all items are entered, the receptionist clicks **Start Testing**.
- The system validates that every item has a test type assigned. If valid, the job moves to the first required test stage and disappears from the Reception Desk.
- It now appears on the relevant lab desk.

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4. How Jobs Move Through Test Stages

4.1 The Test Sequence

- Tests always follow this fixed order: **XRF** → **Fire Assay** → **Titration**.
- But a job only passes through the stages that its items require.

Examples:

Items in the Job	Stages the Job Passes Through
2 gold items needing XRF only	Draft → XRF → Billing → Done
1 gold item needing XRF + 1 gold item needing Fire Assay	Draft → XRF → Fire Assay → Billing → Done
1 gold XRF + 1 silver Titration	Draft → XRF → Titration → Billing → Done
Items needing all three tests	Draft → XRF → Fire Assay → Titration → Billing → Done

4.2 Advancing Between Stages

- At each test stage, the lab operator enters results (see Sections 5–7), then clicks **Approve & Next**.
- The system validates that all required results are complete before advancing. If anything is missing, an error message tells the user exactly what needs to be completed.

4.3 What Gets Validated Before Advancing

Current Stage	What Must Be Complete
XRF	Every XRF item must have readings entered for all its pieces
Fire Assay	Every Fire Assay item must have a result record with all strip and check lines filled
Titration	Every Titration item must have a result record with all sample lines filled

4.4 Result Tabs — What's Visible When

The form shows result tabs that accumulate as the job progresses:

At This Stage	Tabs Visible
XRF	Service Items + XRF Results
Fire Assay	Service Items + XRF Results + Fire Assay Results
Titration	Service Items + XRF Results + Fire Assay Results + Titration Results
Billing / Done	All of the above + Billing Summary

A tab only appears if the job actually has items for that test type. For example, a job with only XRF items will never show Fire Assay or Titration tabs.

4.5 Edit Lock

Once a job leaves Draft, all job details (customer, items, weights, etc.) are locked and cannot be changed. Only the state can advance forward. This prevents accidental modification of test data.

4.6 Cancellation

A job can be cancelled from any state except Done using the **Cancel** button. Cancelled jobs remain in the system for reference.

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5. XRF Testing

5.1 Overview

- XRF (X-Ray Fluorescence) testing measures the percentage composition of metals in an item.
- Each item can have multiple pieces (based on the Qty entered at reception), and each piece gets its own XRF reading.

5.2 Entering XRF Results

1. The XRF operator opens a job from the **XRF Desk**
2. In the Service Items tab, each XRF item has a **"XRF Test"** button
3. Clicking the button opens the **XRF Result Entry** popup, starting at Piece 1

5.3 The XRF Result Entry Popup

Step 1 — Choose a File Template

At the top of the popup, radio buttons allow selecting which type of XRF file to upload. Each template corresponds to a different set of metals the XRF machine reports:

Template	File Type	Number of Metals	Key Metals
ALL (default)	A.EXP	21	Au, Ag, Cu, Zn, Ni, Cd, In, Ir, Ru, Rh, Pd, Pt, Os, Re, Mn, Co, Ga, Sn, Pb, Bi, W
FINE	F.EXP	6	Au, Ag, Cu, Ir, Ru, Os
SILVER	S.EXP	11	Ag, Cu, Ni, Zn, Cd, Pb, Fe, Sn, Ge, Se, Au

HALLMARKING	H.EXP	20	Au, Ag, Cu, Zn, Ni, Cd, In, Ir, Ru, Rh, Pd, Os, Re, W, Fe, Co, Mn, Sn, Pb, Ga
JEWELLERY	J.EXP	15	Au, Ag, Cu, Zn, Ni, Cd, In, Ir, Ru, Rh, Pd, Os, Sn, Pb, Sb

When the template is changed, any previously uploaded file and parsed data are cleared.

The metal columns visible in the results table change based on the selected template — only the metals relevant to that template are shown.

Step 2 — Upload the .EXP File

The user uploads the file from the XRF machine. Only `.EXP` files (A.EXP, F.EXP, S.EXP, H.EXP, J.EXP) are accepted. The system:

- Reads the last 8 rows of the file
- Parses each row into metal percentage columns based on the selected template
- Displays the rows in a table
- Pre-selects the first 2 rows with a checkbox

If the uploaded file's extension doesn't match the selected template, a warning is shown but parsing proceeds using the selected template's column order.

Step 3 — Select Rows and Proceed

The user reviews the parsed rows and uses the checkboxes to select which rows should be averaged. At least one row must be selected.

- If this is **not the last piece**: Click **Next** — the system averages the selected rows, saves the result for this piece, and opens the popup for the next piece
- If this is **the last piece**: Click **Done** — the system averages, saves, and closes the popup

The selected template carries forward to the next piece automatically.

5.4 XRF Results Tab

After results are entered, the **XRF Results** tab on the main job form shows a summary table:

S.No	Item Name	Type	Purity Claimed	Piece	XRF Gold Purity % / XRF Silver Purity %

- For gold items, the "XRF Gold Purity %" column is shown

- For silver items, the "XRF Silver Purity %" column is shown instead
- Each piece appears as a separate row (Piece 1, Piece 2, etc.)

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6. Fire Assay Testing

6.1 Overview

- A sample is drawn from the item, processed through cupellation, and the resulting cornet is weighed to calculate fineness.
- For a lot of identical pieces, one sample is drawn from this lot. So, in the module, there will be one fire-assay result entry for one item even though its quantity can be more than one.

6.2 Entering Fire Assay Results

1. The Fire Assay technician opens a job from the **Fire Assay Desk**
2. In the Service Items tab, each Fire Assay item has a **"Fire Assay Test"** button
3. Clicking the button opens the **Fire Assay Result Entry** popup

6.3 The Fire Assay Popup

Header Section:

Field	Description
Sample Drawn Weight (Mg)	Weight of sample taken from the item
Button Weight (Mg)	Weight of the lead button
Sampling Method	How the sample was taken: Scraping, Cutting, or Micro Drilling
No. of Samples	How many samples were taken (changing this regenerates the lines below)

Sample Lines Table:

When "No. of Samples" is set (e.g., 2), the system automatically creates:

- **Strip lines**: Strip 1, Strip 2 — these are the primary measurements
- **Check Gold lines**: Check 1, Check 2 — these are verification measurements

Each line has these columns:

Column	Strip Lines	Check Lines
Sample	Label (e.g., "Strip 1")	Label (e.g., "Check 1")
M1 (g)	Initial sample weight — enter manually	Enter manually
Silver (g)	Silver weight — enter manually	Enter manually
Copper (g)	Copper weight — enter manually	Enter manually
Lead (g)	Lead weight — enter manually	Enter manually
M2 (g)	Cornet weight after assay — enter manually	Enter manually
Delta	Not shown for strips	Auto-calculated: M1 − M2
Fineness	Auto-calculated (see formula below)	Not shown for checks

How the calculations work:

- **Delta** (on check lines) = M1 − M2
- **Average Delta** = Average of all check line deltas (shown in summary bar)
- **Fineness** (on strip lines) = (M2 + Average Delta) × 1000 ÷ M1
- **Mean Fineness** = Average of all strip fineness values

Summary Bar:

Field	Description
Avg Delta	Average of all check line deltas
Mean Fineness Report (W)	Average fineness from strip lines. Can be manually edited within a small tolerance (default ±0.5, configurable per branch)
Calculated	Shows the auto-calculated mean fineness for reference
Remarks	**Pass** or **Fail** — auto-determined by comparing Mean Fineness against the Purity Claimed

How Pass/Fail is determined: The system reads the Purity Claimed value from the item line and converts it to a fineness threshold:

- "K22" → 916.667 (22/24 × 1000)
- "995" → 995 (already in fineness scale)
- "91.9" → 919 (percentage × 10)

If Mean Fineness ≥ threshold → **Pass**. Otherwise → **Fail**.

Click **Save & Close** to save results.

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7. Silver Titration Testing

7.1 Overview

Silver Titration determines silver purity by dissolving a sample and titrating it with a standard solution. The volume consumed indicates the silver content.

7.2 Entering Titration Results

1. The Titration technician opens a job from the **Silver Titration Desk**
2. In the Service Items tab, each Titration item has a blue **"Titration Test"** button
3. Clicking the button opens the **Titration Result Entry** popup

7.3 The Titration Popup

Header Section (blue background):

Field	Description
No. of Samples	How many samples were taken (changing this regenerates the lines below). Default is 2
Act Norm	The actual normality of the titrating solution — a calibration factor

Sample Lines Table:

When "No. of Samples" is set, the system creates Sample 1, Sample 2, etc.

Column	Description
Sample	Label (e.g., "Sample 1") — auto-generated, read-only
Weight of Sample (mg)	Weight of the dissolved sample — enter manually
EpmL (Volume)	Volume of titrant consumed — enter manually
Fineness	Auto-calculated: (Volume × Act Norm) ÷ Weight of Sample

Summary Bar:

Field	Description
Mean Fineness	Average of all sample fineness values
Remarks	**Pass**, **Fail**, or **Repeat** — auto-determined by comparing Mean Fineness against the Purity Claimed (same threshold logic as Fire Assay)

Click **Save & Close** to save results.

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8. Billing & Completion

8.1 Billing Summary Tab

Once all tests are complete, the job advances to the Billing state. The **Billing Summary** tab shows a consolidated read-only view of all items and their test results:

Column	Description
S.No	Item sequence number
Item	Item name
Type	Item type
Qty	Number of pieces
Wt. Recorded	Recorded weight
Purity	Purity claimed
Test	Which test was performed
XRF Purity (%)	Shown only for XRF items — the XRF purity result
Fire Assay Result	Shown only for Fire Assay items — the mean fineness value
Titration Result	Shown only for Titration items — the mean fineness value

8.2 Completing the Job

From the Billing state, click **Send to Done** to finalise the job. The job moves to Done state and disappears from all active desks. It remains accessible through the **All Jobs** view.

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9. Key Business Rules Summary

Rule	Behaviour
Metal-test combination	Gold: XRF or Fire Assay. Silver: XRF or Titration. Platinum: XRF only
Auto-metal assignment	Selecting Fire Assay → Metal becomes Gold. Selecting Titration → Metal becomes Silver
Weight check	Automatic match/mismatch badge comparing claimed vs recorded weights
XRF file acceptance	Only .EXP files are accepted (5 types: A, F, S, H, J). Template must be selected before upload
XRF row averaging	At least 1 row must be selected. First 2 are pre-selected
Fire Assay edit limit	Mean Fineness can be manually adjusted within ±0.5 (branch-configurable) of the calculated value
Pass/Fail determination	Automatic comparison of test results against claimed purity threshold
Job edit lock	Once confirmed (leaves Draft), no job details can be modified — only state advances
Purity values protected	A purity value in the dropdown cannot be deleted if any item references it
Validation before advancing	System checks that all test results are complete before allowing stage progression

Refinery

Refinery Module - Functional Specification

Overview,
What the Refinery Module Does

Building Blocks of the Refinery Module:

SNO	BLOCK	FUNCTION
1	Intake Desk	Creates a refinery order, captures work type (`branch` or `customer`), refinery centre, raw gold sent weight, cuttings sent, and declared purity. The order reference is auto-generated from the branch short name, date, customer ID, and sequence.
2	Receipt Desk	Captures received gold weight, received cuttings, and melting details. This is the first place where the actual received material is recorded against the intake order.
3	Intake Approval Desk	Compares sent and received weights, computes hand loss, exposes XRF purity handling, and records center approval before orders can be grouped into a refinery batch.
4	Intake Purity Testing	Handles intake-stage XRF in 2 ways: local XRF result entry from the order form when a machine is available, or send-to-branch tracking through (`phrpl.gold.xrf)` when the sample must be tested elsewhere.
5	Refinery Batch Creation	Creates one `phrpl.refinery.stage` from selected centre-approved orders, auto-picks the batch centre from linked orders, auto-names the batch as `BLR_YYYYMMDD_n`, and deducts `Raw Gold` stock from the related refinery centres.
6	Aqua-Regia	Records chemical usage for batch refining. The module also computes suggested HCL, HNO3, HH-Acid, and Urea amounts from total melting weight and warns when entered usage exceeds branch tolerance.
7	Refining & Assay	Records pure gold recovery, pure sample weight, expected pure gold, actual pure gold, and refinery loss. This stage is also where phase-2 purity testing is supposed to happen.

8	Final Processing	Computes prep purity and required added metal weight, deducts added alloy metals from branch inventory, records finished goods output lines, and captures total cuttings created
9	Back to Inventory / Allocation Review	Reviews how final output and cuttings should be distributed back across non-customer orders, creates the downstream `zen.refinery` history record, and consumes acids/materials in inventory.
10	Billing	Creates a refinery billing record from either a gold order or a refinery batch stage, prints a bill, and tracks payment method, payment status, and balance amount

Functionalities in the Module:

1. Order naming is automatic. The current format is
   `BRANCHCODE-YYYYMMDD-CUSTOMERID-SEQ`
   For branch-only work with no customer, the customer part becomes `0000`.
2. Customer work type is enforced at the model level. If (`work_type = customer)`, a customer record is mandatory.
3. Intake validations are active for centre, sent weight, and declared purity. Receipt validates that received gold cannot exceed sent gold and that received cuttings cannot exceed sent cuttings once the record is already in the Receipt stage.
4. Moving from Receipt to Intake Approval requires:
   - The order to be accepted requires
   - a positive melting total weight and
   - a positive melting sample weight.
5. For the branch work type, moving past Receipt adds received raw gold weight into the refinery centre's (`Raw Gold)` stock bridge record.
6. Intake-stage local XRF uses an `.exp` upload wizard:
   - The last 8 lines of the file are parsed,
   - The latest 2 scans are preselected,
   - The selected scans are averaged and written into (`phrpl.gold.xrf.order.result`.)
7. If no XRF machine is available, the module creates a `phrpl.gold.xrf` tracking job and sends the melting sample to another branch.
8. The Testing menu does not represent every XRF intake performed in the module. Local intake XRF results stay on the order-level `phrpl.gold.xrf.order.result`, while the Testing menu mainly shows sent-to-branch and refinery-stage-linked XRF jobs.
9. Aqua-Regia suggested quantities are currently hard-coded from the total batch gold weight:
   - HCL: `1690 ml` per kg

- HNO3: `320 ml` per kg
- HH-Acid: `300 ml` per kg
- Urea: `2000 g` per kg

10. Acid deviation warnings depend on branch acid master tolerances (`acid_loss`). Hand-loss warning depends on the branch `hand_loss`. Refinery-stage gold-loss warning depends on branch `gold_loss`.
11. The expected pure gold for a refinery batch is calculated as:
    `sum(receiving_gold_received_weight * observed_purity / 100)`
12. Prep metal weight is calculated as:
    `((phase_2_purity - branch_market_value) / 100) * pure_gold_weight`
13. Added metal lines are inventory-aware. Creating, editing, or deleting a metal line immediately deducts or restores stock from the selected branch.
14. Finishing a final-processing batch can push `goods_weight + total_cuttings` into a selected refinery inventory product. That pushed quantity is tracked in `inventory_gold_added,` so the later inventory handoff can reverse it cleanly.
15. Back-to-Inventory allocation is proportional. Each eligible order gets a share of refined output and cuttings based on its input share in the batch.
16. Billing records are created with stage data autofilled, but pricing, taxes, and payment amounts are still entered manually in the billing form.

Business Records

Business Records- Overview of all the Ops

Functional Specification

Building Blocks of Business Records Module:

SNO	BLOCK	FUNCTION
1	Monthly Business Record Header	Creates one parent record per `branch + month + year`, stores the monthly identity, state, filing/sending dates, branch location data, center type, and the top-level summary totals shown in list/kanban/form views.
2	Transfer Data Wizard	Takes a branch and date range, finds or creates the monthly record for every affected month, deletes previously transferred lines in that range, then rebuilds the record from live source models.
3	Cash Part	Stores per-day service breakup rows for non-HUID, HUID, card, touch, fire assay, other income, due tracking, and cash-related summary columns in the same sheet structure as the workbook template.
4	HUID Billing	Stores per-day customer billing rows with customer name, GSTIN, total pieces, rejected pieces, service amount, tax breakup, royalty, and total billed amount.
5	Market Due	Stores customer-wise due snapshots using debit and credit values derived from transaction balances and collected amounts.
6	Branch Expenses	Stores branch-level operating expenses such as description, amount, mode, vendor/head, and a few extra bookkeeping columns that match the Excel format.
7	Bank Sheet	Stores date-wise bank payment rows with party name, payment mode, amount, bank name, notes, and TDS-related columns.
8	Refinery Sheet	Provides a manual sheet structure for refinery process, kaccha, recovery, profit, silver movement, and balance figures
9	Corporate Expenses	Stores corporate expense rows with particulars, expense head, type, amount, TDS, and total.
10	Excel Export	Rebuilds the Business Record into an 8-sheet downloadable workbook: `Basic Info`, `Cash part`, `Expenses Details`, `Refinery part`,

		`Bank Sheet`, `HUID Billing Details`, `Market Due List`, and `corporate expenses`
11	Dashboard Aggregation API	Computes date-filtered KPIs such as record counts, cash/bank/expense totals, net total, market due total, day-wise trend series, service revenue mix, aging buckets, and top debtors.
12	Legacy Excel Import / Raw Sheet Archive	Accepts an uploaded workbook, stores the raw file on the parent record, and preserves each imported row as JSON in `business.record.sheet.line` for fallback/reference purposes.

Functionalities in the Module:

17. Record uniqueness is enforced at the SQL level. Only one Business Record can exist for the same `branch_id + month + year`.
18. Record naming is intended to be automatic. The current format is: `BR/<branch_name>/<year>-<month>`
19. `period_display` is computed as `YYYY-MM`, so the monthly identity is readable in list, kanban, pivot, graph, and form views.
20. The transfer wizard supports both single-day and multi-day ranges. If the chosen range crosses month boundaries, it creates or reuses one monthly record for each month touched by that range.
21. Re-running the transfer is intended to be idempotent for date-based lines. Before recreating data, the wizard deletes the already existing cash, expense, corporate, bank, and HUID rows that fall inside the selected dates.
22. Confirmed `transaction.entry` records are the main source for the operational tabs. The module reads billing totals, tax amounts, paid amounts, balance amounts, and piece counts from transaction lines.
23. Cash-part rows are grouped by `(date, customer)`. For each customer-day bucket, the wizard computes non-HUID, HUID, card, touch, and fire-assay amounts using customer-specific rate setup with fallback to default rate setup.
24. HUID Billing rows are generated from the same transaction buckets and carry customer name, GSTIN, total pieces, HUID pieces, rejected pieces, taxable amount, GST breakup, royalty, tax, and total.
25. Market Due rows are grouped customer-wise across the selected range using `balance_amount` as debit and `amount_paid` as credit.
26. Branch expenses are copied from `branch.expense`, corporate expenses are copied from `phrpl.corporate.expense`, and bank-sheet rows are copied from `transaction.payment`.

27. The export button generates a real XLSX file in memory, stores it as an attachment, and returns a direct download URL to the user.
28. The module is editable after transfer. All notebook pages use editable list views, so users can manually complete or correct rows inside the monthly record.
29. `compute_dashboard_data()` is not just a list helper. It returns structured dashboard output including totals, trends, aging, top debtors, and filter options for outside consumers.
30. The revenue mix reported by `compute_dashboard_data()` is split into `non_huid`, `huid`, `refinery`, and `ancillary` buckets based on cash-line and refinery-line values already stored on the monthly records.
31. The legacy Excel import path always preserves a generic raw-sheet row in JSON form, even when the typed import path is incomplete or fails to reflect the actual line-model schema.

Billing

Billing- Overview of Settlement and Transactions

Functional Specification

Building Blocks of Billing Module:

SNO	BLOCK	FUNCTION
1	Customer & Rate Desk	Extends `zen.customer.details` with transaction history, billing ledgers, advances, exchange balance visibility, and discount links. Service pricing is pulled from the customer-specific rate setup first and the default rate setup second.
2	Transaction Entry Desk	Creates the common `transaction.entry` header for all billing types, stores branch/customer/job metadata, controls workflow state, computes service/tax totals, and generates the branch-coded invoice number.
3	Hallmarking Desk	Uses the shared transaction engine with hallmarking-specific delivery metadata, summary totals, and a validation hook that expects at least one billable hallmarking service quantity on the lines.
4	Purity Testing Desk	Reuses the shared transaction engine for testing jobs, sample/report metadata, and testing summaries. It is also the place where testing jobs can be marked as done after billing confirmation.
5	Non-HUID & Miscellaneous Desk	Uses the same line model for non-HUID marking and miscellaneous manual-rate services such as logo, laser, coin work, diamond refinery work, certification cards, and old-gold handling rows.
6	Exchange Desk	Runs the weight-based gold exchange flow: computes fine weight from purity, keeps service billing at zero, tracks gold returned through `exchange.return.line`, and calculates remaining gold and cash liability using `exchange_gold_rate`
7	Payment & Settlement Desk	Registers normal inflow payments (`cash`, `upi`, `bank`, `adjustment`, `gold_physical`) and exchange outflows (`return_cash`, `return_upi`, `return_bank`, `return_gold`). It also auto-posts customer ledger movement on payment creation.
8	Customer Ledger	Maintains a customer-wide financial ledger in `customer.billing.ledger` covering invoice raised, payment

		received, advance deposited, advance consumed, refund, exchange info, exchange cash out, discount applied, and reversals.
9	Branch Ledger	Maintains a branch-side cash and revenue ledger in `branch.billing.ledger` covering revenue, cash movement, advance receipt, inter-branch movement, expenses, scrap receipt, and reversals.
10	Advance Deposit Desk	Captures customer advance deposits, confirms them into the ledgers, auto-applies them FIFO to open invoices of the same customer and same branch, and keeps an `advance.consumption` audit trail.
11	Discount Approval Desk	Captures invoice-level discounts, supports threshold-based approval, and posts approved discounts into the customer and branch financial trail.
12	Branch Expense Desk	Records branch operating expenses with submit/approve/reject workflow and is intended to post approved expense amounts into the branch ledger.
13	Corporate Expense Desk	Records corporate-level expenses with amount, TDS, total, and status, mainly as a separate finance capture area rather than as a live branch-ledger posting flow.
14	Scrap Reconciliation Desk	Tracks daily testing scrap reports and monthly scrap validation by comparing reported scrap with estimated scrap from confirmed exchange and purity-testing activity.
15	Financial Report Wizard	Produces service summary, customer statement, branch performance, ageing, exchange gold flow, and customer comparison style outputs from transactions and ledgers.
16	Complete Records Export	Generates the 7-sheet `Complete Records` workbook with `Cash part`, `Expenses Details`, `Bank Sheet`, `Refinery part`, `HUID Billing Details`, `Market Due List`, and `corporate expenses`.
17	Invoice / Challan Printing	Prints the transaction through a QWeb `Invoice cum Delivery Challan` template with different sections for hallmarking, purity testing, exchange, and other service cases.

Functionalities in the Module:

32. One shared transaction engine handles 5 billing types: `hallmarking`, `exchange`, `purity_testing`, `non_huid`, and `miscellaneous`.
33. Invoice numbering is automatic and branch-coded. The module maps each transaction type to its own sequence and prefixes it with the branch `invoice_code`.
34. The main workflow is strict at the header level:
    `draft -> validated -> confirmed -> cancelled`
    with audit users/datetimes stored for validation and confirmation.
35. Service amount computation is line-driven. For non-exchange transactions, `total_amount` is the sum of `line.amount`, then CGST, SGST, IGST, and incidental charges are added to `service_subtotal`.
36. Exchange records are intentionally not treated as cash invoices. Their `grand_total` stays zero, while the gold and cash settlement math is handled separately from the service-billing path.
37. Rate lookup is customer-first. If a customer-specific rate does not exist for the line product category, the module falls back to the default active rate setup.
38. Confirmation locks line-level service breakup values into dedicated stored fields such as `huid_amount_locked`, `xrf_amount_locked`, `fire_assay_amount_locked`, and `service_total_locked` so later reporting can read the frozen values instead of re-deriving them from changed rates.
39. Confirming a normal invoice auto-consumes available customer advance deposits if the invoice still has a positive `balance_amount`. The consumption is FIFO and branch-isolated.
40. If the transaction is linked to a testing job, confirmation can push that testing job from billing into done status.
41. Exchange lines compute 24K-equivalent `fine_wt` from gross/net weight plus declared purity. The header then aggregates total fine weight, fine-weight balance, cash paid out, cash collected back, and the net cash balance still owed either by the branch or by the customer.
42. Exchange settlement supports three distinct completion paths:
    - return gold physically,
    - pay the customer cash/bank/UPI for the remaining liability,
    - Collect surplus money from the customer if the gold received exceeded the obligation.

12. Creating a payment auto-posts the customer ledger entry. `gold_physical` tender additionally creates a `gold.scrap.log` row, so gold-as-payment is not only a cash event.
13. The module uses two ledgers, not one:
    - `customer.billing.ledger` for the customer's financial position across branches
    - `branch.billing.ledger` for branch-side revenue/cash/expense posting

14. Advance deposits are operational, not passive. Confirmation posts the deposit, auto-applies it to open same-branch invoices where possible, and marks the deposit exhausted when fully consumed.
15. Discount records support draft, pending approval, approved, and rejected states with branch-level approval threshold logic and a dedicated discount reference sequence.
16. Branch expenses and corporate expenses are both captured inside the module, but they are not the same thing: branch expenses are tied to branch operations, while corporate expenses are a separate corporate log with TDS-aware totals.
17. Scrap management is not a placeholder. Daily scrap reports roll into monthly validation records that compare expected scrap against actual scrap and compute weighted average purity and pure-gold variance.
18. The report wizard is not just a print button. It builds transient analysis lines and summary KPIs for service mix, ledger-style customer views, branch performance, ageing, exchange gold movement, and customer comparison.
19. The `Complete Records` export wizard rebuilds a workbook directly in memory, stores it as an attachment, and returns a direct download action instead of only rendering an on-screen list.
20. The module includes a real, printable transaction report layer, but the print template is only as accurate as the live model fields it references.

Exchange

Exchange Module - Functional Specification

Exchange Module is used for implementing the exchange process which is a key service of PHRPL. It is inter-connected to some of the processes like it is connected to a refinery.

Exchange Module will be having two main features:

Exchange Process

Exchange Return

Exchange Process: The process flow of exchange happens like this--->

Customer_details → customer_product_details(katcha) → testing → pure gold calculation → pure gold exchange to the customer → done.

This process is followed by the PHRPL team, so i am trying to recreate that project in our exchange module.

Customer Onboarding/ Customer details: This is the first step for the customer onboarding for the exchange process. We need some details from the customers like name, address, phone no etc.

Once we select the customer these fields will be auto filled. If The customer is new , we have an option for creating customer from there it self.

These customer creation is possible because we are connecting masters module.

Fields Used:

1. Name
2. Phone
3. Address

4. Date
5. Centre(branch)

Customer Product details: This is one of the main step where the exchange process starts. Customers will come to us with some of their products/items like gold katcha. These product details we need for checking if the customer is claiming right details. Customer walks in claims the weight of product and purity of the product those things we have to check.

Fields Used:

1. Item Name(product)
2. Claimed Weight
3. Claimed Purity
4. Observed Weight (This is the main thing we have to check)

Testing: This is one of the main parts of exchange process. Here the product which is brought by the customer should be tested by using the XRF machine. The results are validated and pure weight we have to return to the customer will be directly calculated.

This state or this phase is only available for the tester from the employees. So once you install the module you have to give the groups to your employees.

Fields Used:

1. Gold purity results
2. Silver purity results
3. Other metal purity values.
4. wastage when scraped for testing.

Pure Gold calculation: This is automatically calculated by using the results of the testing.

Pure weight should be given and Purity.

Final Settlement and change in purity:

This step or stage is for addition of cash paid and changes in purity observed by the xrf machine.

This is used in phrpl exchange process.

Fields Used:

1. Extra (it is a boolean basically by clicking it on then we are saying that we are getting cash for selling extra gold. if it is off we are saying that we are giving cash for exchange less gold)
2. requested gold amount
3. exchange rate (this should be coming from api call)
4. cash payable (cash should be paid to the customer)
5. cash received (cash should be paid by the customer)
6. payment type(dropdown which shows that which type of payment)
7. amount (this is finally calculated amount)
8. Item id(this is also selection but gives the items which are in line)
9. add/ subtract buttons to increase or reduce purity.

Retest: This is mainly a feature where after testing customer is not satisfied with the purity results. Then we can ask for retesting.

Exchange return:

This step is added because some cases might raise where already exchange is in done state but the customer wants a return then it should not be closed just by cancelling. It should raise a exchange return which will need approval from the admin office for returning exchange.

Fields used:

1. original exchange will be showing all the exchanges. You can select any exchange you want.

for now it is returning directly we should a raise a request method for exchange return.

Reports: The reports contain the accumulation of all the data which will be filtered through the filters we have given different types of report. Daily report, Monthly report, Customer wise report.

Daily report is basically will have a filter based on days and monthly report will have a filter based on months, customer report will have a filter based on customer.

SNO	BLOCK	FUNCTION
1	Exchange Registry	An Exchange registry is created here for the customer onboarding and item details then passed to the testing stage.
2	Testing Stage	An Exchange Testing is a stage where testing for the onboarded customer and his item are done. There will be a testing button which can be clicked and can enter the results.
3	Final Review Stage	This stage is for the final review where the user can check the details of the testing and also he will have a chance to change some values that are related to the masters module. After this it updates the inventory and receipts and deliveries.
4	Done Stage	This stage is the final stage where you can print the receipt and also this is where it goes to the billing module and creates a record in that module.
5	Cancelled Stage	This is for cancelling in any stage except the done stage. Because in the done stage it updates the inventory and by cancelling directly we cannot directly subtract the inventory.
6	Exchange Return	This is the main process for the return process. This comes into picture when the exchange process is done and the customer wants to return the pure gold.
7	Daily Report	This is a report generating screen. This works basically based on start date and end date it gives all the records of exchange transactions in between dates.
8	Customer Report	This is a report generating screen. This works basically based on the customer. It gives all the records of exchange transactions belonging to that customer.
9	Item Report	This is a report generating screen. This works basically based on the

		Item/Product. It gives all the records of exchange transactions belonging to that Item/Product.
10	Branch Report	This is a report generating screen. This works basically based on the Branch/Centre. It gives all the records of exchange transactions belonging that Branch/Centre.

Laser Processes

Laser Marking Module — Functional Specification

1. What This Module Does

- The Laser Marking module manages the end-to-end process of laser processes on the items.
- A customer brings items (gold, silver, or platinum) for laser marking — this can include non-HUID engraving, seal marking, or logo marking.
- The module tracks the order from reception through marking to completion,
- generates transaction entries for billing,
- provides a daily validation dashboard that reconciles the number of markings done against the laser machine's part counter.

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2. Desks & Menu Structure

S.No	Menu Item	Who Uses It	What They See
1	Reception Desk	Receptionist	All jobs in Draft state — new orders being entered
2	Marking Desk	Laser Operator	Only jobs in Marking state — items ready to be marked
3	All Orders	Manager / Admin	Every job regardless of state
4	Validation	Supervisor	Daily dashboard reconciling marking counts vs machine counter
5	Reports	Supervisor	Upload area for machine .markcfg files

---

3. Creating a Laser Marking Job

3.1 Starting a New Order

The receptionist opens the Reception **Desk** and clicks **New**. A blank order form opens.

3.2 Order Details

Order ID is displayed prominently at the top in the format `WH-HM-XXXX-YYY` where XXXX is a sequential number and YYY is the total marking count. This ID updates automatically as markings are recorded.

Service Details:

Field	How It Works
Service	Pre-filled as "Laser Marking" (read-only)
Metal Type	Select Gold, Silver, or Platinum (required)
Date	Auto-filled with current date and time
Branch	Select the branch where marking is done
Type of Marking	Select the marking type (e.g., Non-HUID Engraving, Seal, Logo). New types can be added from the dropdown

Customer Information:

Field	How It Works
Customer Name	Select from the customer list. All other fields auto-fill
Customer Number	Auto-filled from customer record
Customer Company	Auto-filled
Customer GSTIN	Auto-filled
Customer BIS Number	Auto-filled
Customer Address	Auto-filled

3.3 Adding Items

The **Items** tab contains the items to be marked:

Column	How It Works
S.No	Auto-numbered sequentially

Item	Select the product (e.g., "Ring", "Chain") — required
Quantity	Number of pieces to mark. Default is 1
Type	Item type category (defaults to "Type 1"). New types can be added
Type of Marking	Marking type for this specific item. Overrides the job-level marking type if set. New types can be added
Weight per item	Weight of each piece in grams
Total Weight	Auto-calculated: Quantity × Weight per item

Remarking behaviour:

- Each item line tracks a **remarking count** (minimum 1, cannot be zero)
- The **Total Marking** count shown on the job is the sum of all remarking counts across items

Same-item logic:

- When the same item name is added more than once, the Type field is left empty for the user to choose
- When an item name is added for the first time, it defaults to "Type 1"

3.4 Saving the Order

The order is saved in **Draft** state and remains visible in the Reception Desk.

---

4. Marking Process

4.1 Starting Marking

When the receptionist clicks **Start Marking**, the system:

1. Generates individual **marking lines** — one per piece. For example, if Item A has Qty 3 and Item B has Qty 2, the system creates 5 marking lines: "Piece 1", "Piece 2", "Piece 3" for Item A and "Piece 1", "Piece 2" for Item B
2. Moves the job to **Marking** state
3. The job now appears on the **Marking Desk** and disappears from the Reception Desk

4.2 The Marking Tab

The **Marking** tab becomes visible only when the job is in Marking state. It shows the auto-generated per-piece lines:

Column	How It Works
S.No	Auto-numbered
Item	The item name (from the Items tab)
Piece	"Piece 1", "Piece 2", etc.
Type	Item type (carried from Items tab, can be changed)
Is Remarking	Checkbox — flag if this piece needs remarking
Total Remarked	Number of times this piece was marked. Default is 1

Total Markings is displayed at the bottom of the tab — the sum of all "Total Remarked" values across all marking lines.

4.3 Completing the Job

When the operator clicks **Marking Completed**, the system:

1. Creates a **transaction entry** for billing with the following details:
    - Transaction type: Non-HUID
    - Branch, date, and customer from the job
    - One transaction line per item with product, quantity, gross weight, net weight, metal category, and marking type quantities (non-HUID, seal, or logo based on the marking type name)
2. Moves the job to **Done** state

How marking type determines transaction quantities:

- If the marking type name contains "engrav", "marking", or "non" → counted as **Non-HUID**
- If the marking type name contains "seal" → counted as **Seal**
- If the marking type name contains "logo" → counted as **Logo**
- If none of these match → defaults to **Non-HUID**

4.4 Done State

When the job is done, a green ✓ **Marking Completed** button is displayed as a visual confirmation. The job remains accessible through **All Orders**.

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5. Reports — Machine File Upload

5.1 What Reports Are For

At the end of each working day (or shift), the laser machine exports a `.markcfg` configuration file. This file contains a running counter of total parts marked by the machine. The Reports section captures this data and computes how many markings were done since the last upload.

5.2 Creating a Report

Open **Reports** from the menu and click **New**.

Field	How It Works
Report ID	Auto-generated: `MR-{date}-{sequence}` (e.g., `MR-20260428-001`)
Date	Auto-filled with current date and time
Upload File	Upload the `.markcfg` file from the laser machine (required)

5.3 What Happens on Upload

When the file is uploaded, the system:

1. **Parses** the file and extracts the `TOTALPARTNUM` value (the machine's cumulative parts counter)
2. **Looks up** the previous report's part number
3. **Calculates the difference**: Current Part Number − Previous Part Number = parts marked since last report

Field	Description
Previous Part Number	The part number from the last uploaded report

Current Part Number	The `TOTALPARTNUM` extracted from the uploaded file
Difference	Current − Previous = how many parts were marked in this period

5.4 Automatic Push to Validation

After saving, the report automatically pushes its **Difference** value into today's Validation record as the **Total Part Number**. This allows the Validation dashboard to compare software-counted markings against machine-counted markings.

---

6. Validation — Daily Reconciliation Dashboard

6.1 What Validation Does

The Validation screen is a daily reconciliation tool. It answers: **"Do our software records match what the machine actually marked?"** Any discrepancy (the Difference value) indicates missed entries, re-markings not recorded, or machine miscounts.

6.2 Opening Validation

Click **Validation** in the menu. The system automatically opens the latest validation record for today. If none exists, it creates one. On opening, it also refreshes all counts.

6.3 The Dashboard

The Validation form shows four colour-coded cards across the top:

Card	Colour	What It Shows	Source
Non-HUID Markings	Orange	Count of markings from today's laser jobs	Sum of remarking counts from all laser job marking lines dated today
HUID Markings	Blue	Count of markings from today's completed hallmarking orders	Sum of total order markings from completed hallmarking orders dated today
Total Markings	Green	Non-HUID + HUID combined	Calculated

Total Part Number	Red	Machine's parts count for today	Pushed from the latest Report's Difference value

6.4 The Comparison Section

Below the cards, a **Comparison** section shows the reconciliation:

Total Markings − Total Part Number = Difference
(green) (red) (blue)

- **Difference = 0**: Software records perfectly match the machine counter
- **Difference > 0**: More markings recorded in software than the machine counted (possible duplicate entries)
- **Difference < 0**: Machine marked more than recorded in software (possible missed entries)

6.5 Refreshing

Click the **Refresh** button in the header to:

- Re-pull the latest Report's difference value
- Recount today's laser job markings and hallmarking order markings
- Recalculate the comparison

The date field updates to the current time on each refresh.

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7. Key Business Rules Summary

Rule	Behaviour
Metal type required	Every job must have Gold, Silver, or Platinum selected before saving
Minimum remarking count	Remarking count cannot be 0 — minimum is 1. System shows a warning and resets to 1
Marking lines auto-generated	Clicking "Start Marking" creates one marking line per piece per item. Existing marking lines are replaced

Order ID updates dynamically	The last 3 digits of the Order ID reflect the total marking count and update on save
Transaction on completion	"Marking Completed" creates a transaction entry for billing based on marking types
Report pushes to Validation	Uploading a .markcfg report automatically updates today's Validation record
Validation auto-creates	Opening the Validation menu auto-creates today's record if none exists
Daily scope	Validation counts are scoped to today's date — only today's laser jobs and hallmarking orders contribute

Inventory

Inventory Module - Functional Specification

1. Overview

- Manages the complete lifecycle of gold and precious metal inventory across branches.
- Tracks receipts, deliveries, internal/external transfers, refinery processing, and generates analytics.

2. Centres (Branch Management)

- Represents physical branch locations where stock is tracked.
- Uses 2–5 letter short codes (e.g., DTH) to generate unique reference numbers.
- Centres can be designated as Main or Refinery. Refinery centres enforce a maximum transfer capacity limit.

3. Products

- The master catalogue of materials, including raw gold, pure gold, and chemicals.
- Stock quantities are tracked individually per branch.
- The system blocks outbound transactions if stock is insufficient.

4. Receipts (Inbound Inventory)

- Records all inbound gold and products into a centre.
- Automatically calculates Pure Weight from Observed Weight and Purity (e.g., 100g at 91.6% = 91.6g pure).
- Multiple receipts can be grouped and sent to a refinery in one action.

5. Deliveries (Outbound Inventory)

- Records all outbound gold to customers, branches, or refineries.
- Tracks gross weight, purity, and pure weight for each item.
- Deliveries for internal or refinery transfers are generated automatically.

6. Refinery Processing

- Tracks step-by-step internal chemical refining for gold batches.
- Logs processing stages like Aqua-Regia, Pure Gold, and Phase-2 XRF.
- Automatically deducts consumed chemicals (HCl, HNO_3, Urea, HH Acid) directly from live branch inventory.
- Monitors input weight, expected output, actual pure weight, and cuttings.

7. Refinery Transfers

- Manages the dispatch of raw gold to external refineries and the return of pure gold.
- Automatically calculates Profit & Loss: P&L Weight = Weight Received − (Weight Sent × Avg Purity / 100).
- Automates the creation of Delivery and Receipt records during the workflow.

8. Internal Transfers

- Handles branch-to-branch movement of goods from request to receipt.
- Auto-calculates net pure weight based on purity tiers (e.g., 22K).
- Uses two-step stock validation: checks availability upon acceptance, and deducts stock upon dispatch.

9. Reports & Analytics

- **Day-Wise:** Summarizes total inflows, outflows, and net movements by date, with click-through drill-down to specific records.
- **Product-Wise:** Tracks individual product line movements (gross and pure weight) across receipts and deliveries.
- **Service-Wise:** Groups inventory movements specifically by Exchange or Refinery transactions.
- **Refinery P&L:** Provides a clear "Gain," "Loss," or "Break Even" verdict on completed refinery transfers.

SNO	BLOCK	FUNCTION
1	Centres (Branch Management)	Manages the physical branch locations where all inventory transactions are tied. It assigns short codes for references, defines if a centre is a main or refinery branch, and sets maximum transfer capacities.
2	Products	Manages the master list of all metals and materials handled in the system. It tracks stock quantities individually at the

		branch level and configures items for sales, purchases, or expenses.
3	Receipts (Inbound Inventory)	Records all inbound movement of gold and products into a centre. It automatically calculates pure weight from the observed weight and purity, and allows users to batch receipts to send to the refinery.
4	Deliveries (Outbound Inventory)	Captures all outbound movement of inventory from a centre, whether to a customer, another branch, or a refinery. It automatically pre-fills details when triggered by internal or refinery transfers.
5	Refinery Processing	Tracks the step-by-step internal chemical refining process for batches of gold. It logs the usage of chemical reagents, monitors expected versus actual output, and tracks cuttings.
6	Refinery Transfers	Manages the workflow of dispatching raw gold to an external refinery and receiving pure gold back. It automatically creates the necessary delivery and receipt records and calculates the Profit & Loss (gain or loss) from the process.
7	Internal Transfers	Handles the movement of stock between internal branches using an end-to-end workflow (request, accept, in transit, received). It includes automatic net weight calculations and stock availability validation before dispatch.

8	Day-Wise Inventory Report	Generates a complete picture of all inventory activity (inflows, outflows, and transfers) for a specific date or date range. It includes a drill-down feature to view underlying records.
9	Product-Wise Inventory Report	Analyzes the inventory movement at the individual product line level. It tracks the total gross and pure weights received and sent out for specific products across selected dates and centres.
10	Service-Wise Report	Displays inventory movements grouped by specific service types, namely Exchange or Refinery. This helps compare the volume and weight of business across these different service lines.
11	Refinery Day-End P&L Report	Provides a financial performance review of refinery transfers for a given date range. It calculates the total P&L weight and cuttings, giving a clear verdict of gain, loss, or break-even based on completed transfers.

Admin/Masters

Admin Module - Functional Specification

1. Employee Management

- **Comprehensive Profiles:** Manages complete employee records, including job titles, departments, employment types (Full-time, Part-time, Contract), and emergency contacts.
- **Automated System Access:** Automatically provisions system user accounts and generates passwords when a new employee is added, linking them directly to their specific system access rights.
- **Financial & Administrative Tracking:** Logs joining and termination dates, salary structures, and allows linking multiple bank accounts to a single employee.
- **Data Integrity:** Automatically formats names and enforces strict rules (e.g., exactly 10 digits for phone numbers, valid email structures) to prevent duplicate or messy data.

2. Customer Management & Rate Setup

- **Customer Profiles:** Stores business details, including Company Name, Contact Info, multiple Bank Accounts, GSTIN, and BIS numbers.
- **Custom Rate Configuration:** Administrators can set specific, approved service charges (Hallmarking, Marking, Seal, XRF, Fire Assay, etc.) tailored to individual customers for Gold, Silver, or Platinum.
- **Rate Approval Workflow:** Features a built-in approval pipeline. Branches can submit "Rate Requests" for specific customers, which management can review, adjust, approve, or reject.
- **Default Rates:** Allows administrators to define system-wide default rates, which can be bulk-applied to customers lacking custom pricing.

3. Branch (Centre) Operations

- **Hierarchy & Categorization:** Manages different physical locations, categorizing them as Main, Testing, Refinery, or Offsite branches. Sub-branches can be linked to a Parent branch for reporting.
- **Operational Thresholds:** Defines operational rules per branch, such as maximum allowed gold capacity, maximum purity adjustments, and expected gold/acid loss metrics.
- **Branch-Specific Inventory & Pricing:** Allows the system to track product stock locally at each branch. Administrators can set minimum/maximum stock alerts and override global product prices with local branch-specific prices.
- **Compliance Validation:** Ensures compliance by validating GST formats, PAN formats, and PIN codes, while automatically keeping Odoo's internal accounting companies in sync with branch data.

4. Product & Master Catalog

- **Centralized Catalog:** Acts as the master list for all items, defining whether they are Goods, Services, or Combos, and tagging them for Sales, Purchasing, or Expenses.
- **Global vs. Local Tracking:** Displays the total combined quantity of a product across the entire business, while also allowing a drill-down into exactly how much stock is sitting at each specific branch.
- **Purchase Tracking:** Logs detailed purchase histories for items (e.g., chemicals or raw materials), tracking the vendor, purchasing branch, quantity, unit price, and discounts, while auto-calculating the final price.
- **Duplicate Prevention:** Uses smart name normalization (ignoring extra spaces and casing) to prevent users from accidentally creating duplicate products.

SNO	BLOCK	FUNCTION
1	Employee Profiles	Manages complete employee records, including job titles, departments, employment types (Full-time, Part-time, Contract), and emergency contacts.
2	Automated System Access	Automatically provisions system user accounts and generates passwords when a new employee is added, linking them to specific access rights.
3	Financial & Administrative Tracking	Logs joining and termination dates, salary structures, and allows linking multiple bank accounts to a single employee.
4	Employee Data Integrity	Automatically formats names and enforces strict rules (e.g., 10-digit phone numbers, valid emails) to prevent duplicate or messy data.

5	Customer Profiles	Stores essential business details, including Company Name, Contact Info, multiple Bank Accounts, GSTIN, and BIS numbers.
6	Custom Rate Configuration	Allows administrators to set specific, approved service charges (Hallmarking, Marking, Seal, XRF, Fire Assay, etc.) tailored to individual customers.
7	Rate Approval Workflow	A built-in pipeline where branches submit "Rate Requests" for specific customers, which management can then review, adjust, approve, or reject.
8	Default Rates Configuration	Defines system-wide default rates that can be bulk-applied to customers who do not have custom pricing set up.
9	Branch Hierarchy & Categorization	Manages physical locations by categorizing them as Main, Testing, Refinery, or Offsite branches, and linking sub-branches to a Parent branch.
10	Branch Operational Thresholds	Defines operational rules per branch, such as maximum allowed gold capacity, maximum purity adjustments, and expected gold/acid loss metrics.
11	Branch Inventory & Pricing	Tracks product stock locally at each branch, sets minimum/maximum stock alerts, and allows overriding global product prices with local branch prices.

12	Compliance Validation	Ensures compliance by validating GST formats, PAN formats, and PIN codes, while automatically keeping internal accounting companies in sync with branch data.
13	Centralized Product Catalog	Acts as the master list for all items, defining whether they are Goods, Services, or Combos, and tagging them for Sales, Purchasing, or Expenses.
14	Global vs. Local Stock Tracking	Displays the total combined quantity of a product across the entire business, while allowing a drill-down into exactly how much stock is at each specific branch.
15	Purchase Tracking	Logs detailed purchase histories for items, tracking the vendor, purchasing branch, quantity, unit price, and discounts, while auto-calculating the final price.
16	Duplicate Prevention	Uses smart name normalization (ignoring extra spaces and casing) to prevent users from accidentally creating duplicate master products.

Notifications

Notifications Module — Functional Specification

A. Overview

1. What this module does

The Notifications module automatically sends alerts when specific actions happen in the system (for example, order received, testing completed, etc.).

Instead of manually informing people, the system:

- Detects a change (like status update)
- Checks rules
- Sends messages to selected users

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2. How it works (simple flow)

1. A record changes state (example: Delivery → Received)
2. System checks matching notification rules
3. If condition matches → notification is triggered
4. Message is sent to selected recipients

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3. Where notifications are used

This module is integrated with:

- Delivery Receipt
- Hallmarking Request
- Other process stages (via system rules)

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4. Types of notifications

- Email notifications
- Telegram notifications
- System-based alerts (future use)

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5. Key concept

Everything is controlled using **Rules**.

👉 No rule = no notification
👉 Wrong rule = wrong message

B. MODULE STRUCTURE

The module has 3 main sections:

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1. Activities

Defines **what event you are tracking**

Examples:

- Delivery Received
- Request Created
- XRF Completed

Each activity is linked to a model (like delivery.receipt or hm.request)

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2. Recipients

Defines **who will receive notifications**

Each recipient can have:

- Name
- Email
- Telegram Chat ID
- Linked system user

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3. Rules

Defines **when and what to send**

A rule connects:

- Activity (what event)
- Trigger (when it happens)
- Recipients (who gets it)
- Message (what is sent)

C. FORMS

1. Activity Form

Purpose: Define event type

Fields:

- Name → Activity name
- Model → System model (example: Delivery, Request)

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2. Recipient Form

Purpose: Define notification receivers

Mandatory / Important:

- Name
- User (optional but recommended)

Communication fields:

- Email → for email notifications
- Telegram Chat ID → for Telegram messages

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3. Notification Rule Form

This is the **main configuration screen**

Basic Section

Fields:

- Name → Rule name
- Activity → Event type
- Model → Auto-filled
- Available States → Shows valid states
- Trigger Value → State at which notification should trigger

Message Section

Default Message

- Auto-generated by system
- Based on record details

Custom Message (optional)

- Enable “Use Custom Message”
- Write your own message

You can insert variables like:

- Customer
- Order ID
- Status

Recipients Section

Fields:

- Recipient list (multiple allowed)